

第二章 习题

第5题：设 R 是集合 A 上的二元关系。定义 A 上的二元关系 R^{op} 使得：

$$xRy \Leftrightarrow yR^{op}x$$

(i) R 给出 A 上的全序结构 $\Leftrightarrow R^{op}$ 给出 A 上的全序结构。

Proof: $(\Rightarrow)$: 对 $\forall x \in A$, $\therefore xRx \therefore xR^{op}x \therefore R^{op}$ 满足反身性。

对 $\forall x, y, z \in A$. 若 $xR^{op}y$ 且 $yR^{op}z$, 则有:

$$\therefore xR^{op}y \quad \therefore yRx \quad \therefore yR^{op}z \quad \therefore zRy$$

$$\therefore zRy \text{ 且 } yRx \quad \therefore zRx \quad \therefore xR^{op}z \quad \therefore R^{op} \text{ 满足传递性.}$$

对 $\forall x, y \in A$. 若 $xR^{op}y$ 且 $yR^{op}x$, 则有:

$$\therefore xR^{op}y \quad \therefore yRx \quad \therefore yR^{op}x \quad \therefore xRy$$

$$\therefore xRy \text{ 且 } yRx \quad \therefore x=y \quad \therefore R^{op} \text{ 满足反称性.}$$

对 $\forall x, y \in A$. $\therefore R$ 是 A 上的全序 $\therefore xRy$ 或 yRx 至少有一个成立

$\therefore xR^{op}y$ 或 $yR^{op}x$ 至少有一个成立

$\therefore (A, R^{op})$ 是全序集.

$(\Leftarrow)$: 与 $(\Rightarrow)$ 方向的证明完全一样. □

(ii) 对于任意集合 S , 给出从偏序集 $(P(S), \subseteq)$ 到 $(P(S), \subseteq^{op})$ 的同构.

Proof: 由 ZFC 幂集公理, $P(S) = \{B : B \subseteq S\}$.

定义映射 $f: P(S) \longrightarrow P(S)$ 下证: f 是严格保序的双射.

$$B \longmapsto S \setminus B$$

对 $\forall B \in P(S)$, 有: $B \subseteq S \quad \therefore f(B) = S \setminus B \in P(S)$

$$\therefore f(P(S)) \subseteq P(S)$$

对 $\forall B_1, B_2 \in P(S)$,

若 $B_1 = B_2$, 则有: $f(B_1) = S \setminus B_1 = S \setminus B_2 = f(B_2) \quad \therefore f$ 是映射.

若 $f(B_1) = f(B_2)$, 则有: $S \setminus B_1 = S \setminus B_2 \quad \therefore B_1 \subseteq S$ 且 $B_2 \subseteq S$

$$\therefore B_1 = S \setminus (S \setminus B_1) = S \setminus (S \setminus B_2) = B_2 \quad \therefore f \text{ 是单射.}$$

对 $\forall T \in P(S)$, 有: $T \subseteq S \quad \therefore S \setminus T \subseteq S \quad \therefore S \setminus T \in P(S)$

$$f(S \setminus T) = S \setminus (S \setminus T) = T \quad \therefore f \text{ 是满射.} \quad \therefore f \text{ 是双射.}$$

对 $\forall B_1, B_2 \in P(S)$,

若 $B_1 \subseteq B_2$, 则有: $S \setminus B_2 \subseteq S \setminus B_1 \quad \therefore f(B_2) \subseteq f(B_1)$

$$\therefore f(B_1) \subseteq^{\text{op}} f(B_2).$$

若 $f(B_1) \subseteq^{\text{op}} f(B_2)$, 则有: $f(B_2) \subseteq f(B_1) \quad \therefore S \setminus B_2 \subseteq S \setminus B_1$

$$\therefore B_1 \subseteq B_2$$

(对 $\forall x \in B_1$, $\therefore B_1 \in P(S) \quad \therefore B_1 \subseteq S \quad \therefore x \in S$.

假设 $x \notin B_2$, 则有: $x \in S \setminus B_2 \subseteq S \setminus B_1 \quad \therefore x \notin B_1$ 矛盾. $\therefore x \in B_2$

$$\therefore B_1 \subseteq B_2)$$

∴ 对 $\forall B_1, B_2 \in P(S)$, 有: $B_1 \subseteq B_2 \Leftrightarrow f(B_1) \subseteq^{\text{op}} f(B_2)$

∴ f 是严格保序的双射.

∴ f 是 $(P(S), \subseteq) \longrightarrow (P(S), \subseteq^{\text{op}})$ 的同构.



S进制展开的唯一性

Lemma: 任取 $s \in \mathbb{Z}_{\geq 2}$, $\forall t \in \mathbb{Z}_{\geq 0}$, ~~$\forall c_0, c_1, c_2, \dots, c_t \in \{-(s-1), -(s-2), \dots, -1, 0, 1, \dots, s-2, s-1\}$~~

$\forall c_0, c_1, c_2, \dots, c_t \in \{-(s-1), -(s-2), \dots, -1, 0, 1, \dots, s-2, s-1\}$

若 $c_0 + c_1 s + c_2 s^2 + \dots + c_t s^t = 0$, 则有: $c_0 = c_1 = c_2 = \dots = c_t = 0$

Proof: 先证明 $c_t = 0$.

假设 $c_t > 0$, 则有: $c_t \geq 1 \quad \therefore c_t s^t \geq s^t$

$\therefore c_0 \geq -(s-1), \quad c_1 s \geq -(s-1)s, \quad c_2 s^2 \geq -(s-1)s^2, \dots,$

$$c_{t-1} s^{t-1} \geq -(s-1)s^{t-1}$$

$$\begin{aligned} \therefore c_0 + c_1 s + c_2 s^2 + \dots + c_{t-1} s^{t-1} &\geq -(s-1)(1 + s + s^2 + \dots + s^{t-1}) \\ &= -(s-1) \frac{1-s^t}{1-s} = 1-s^t \end{aligned}$$

$\therefore c_0 + c_1 s + c_2 s^2 + \dots + c_{t-1} s^{t-1} + c_t s^t \geq 1-s^t + s^t = 1 > 0$. 矛盾.

假设 $c_t < 0$. 则有: $c_t \leq -1 \quad \therefore c_t s^t \leq -s^t$

$\therefore c_0 \leq s-1, \quad c_1 s \leq (s-1)s, \quad c_2 s^2 \leq (s-1)s^2, \dots,$

$$c_{t-1} s^{t-1} \leq (s-1)s^{t-1}$$

$$\begin{aligned} \therefore c_0 + c_1 s + c_2 s^2 + \dots + c_{t-1} s^{t-1} &\leq (s-1)(1 + s + s^2 + \dots + s^{t-1}) \\ &= (s-1) \frac{1-s^t}{1-s} = s^t - 1 \end{aligned}$$

$\therefore c_0 + c_1 s + c_2 s^2 + \dots + c_{t-1} s^{t-1} + c_t s^t \leq s^t - 1 - s^t = -1 < 0$. 矛盾.

$$\therefore c_t = 0. \quad \therefore c_0 + c_1 s + c_2 s^2 + \dots + c_{t-1} s^{t-1} = 0.$$

完全一样的方法，可以证明 $C_{t-1} = \dots = C_2 = C_1 = C_0 = 0$.

$$\therefore C_0 = C_1 = C_2 = \dots = C_{t-1} = C_t = 0 \quad \square$$

Lemma: 任取 $s \in \mathbb{Z}_{\geq 2}$ 又 $\forall n \in \mathbb{Z}_{\geq 0}$ 都有唯一的 s 进制展开:

$$n = a_0 + a_1 s + a_2 s^2 + \dots, \quad a_k \in \{0, 1, \dots, s-1\}$$

其中至多仅有有限个 a_k 非零.

Proof: 存在性已证. 下证唯一性.

假设 $n \in \mathbb{Z}_{\geq 0}$ 有 2 个 s 进制展开:

$$n = a_0 + a_1 s + a_2 s^2 + \dots + a_\lambda s^\lambda, \quad a_k \in \{0, 1, \dots, s-1\}$$

$$n = b_0 + b_1 s + b_2 s^2 + \dots + b_\mu s^\mu, \quad b_k \in \{0, 1, \dots, s-1\}$$

分情况讨论如下:

① $\lambda = \mu$. 则有: $a_0 + a_1 s + a_2 s^2 + \dots + a_\lambda s^\lambda = b_0 + b_1 s + b_2 s^2 + \dots + b_\lambda s^\lambda$

$$\therefore (a_0 - b_0) + (a_1 - b_1)s + (a_2 - b_2)s^2 + \dots + (a_\lambda - b_\lambda)s^\lambda = 0$$

$$\therefore a_k - b_k \in \{-(s-1), -(s-2), \dots, -1, 0, 1, \dots, s-2, s-1\}$$

$$\therefore a_0 - b_0 = a_1 - b_1 = a_2 - b_2 = \dots = a_\lambda - b_\lambda = 0$$

$$\therefore a_0 = b_0, a_1 = b_1, a_2 = b_2, \dots, a_\lambda = b_\lambda$$

② $\lambda > \mu$. $\therefore \lambda \geq \mu + 1$.

$$\therefore a_0 + a_1 s + a_2 s^2 + \dots + a_\mu s^\mu + a_{\mu+1} s^{\mu+1} + \dots + a_\lambda s^\lambda = b_0 + b_1 s + b_2 s^2 + \dots + b_\mu s^\mu$$

$$\therefore (a_0 - b_0) + (a_1 - b_1)s + (a_2 - b_2)s^2 + \dots + (a_\mu - b_\mu)s^\mu + a_{\mu+1} s^{\mu+1} + \dots + a_\lambda s^\lambda = 0$$

$$\therefore a_0 = b_0, a_1 = b_1, a_2 = b_2, \dots, a_\mu = b_\mu. \quad a_{\mu+1} = \dots = a_\lambda = 0$$

$$\textcircled{3} \quad \lambda < \mu \quad \therefore \lambda + 1 \leq \mu$$

$$\therefore a_0 + a_1 s + a_2 s^2 + \cdots + a_\lambda s^\lambda = b_0 + b_1 s + b_2 s^2 + \cdots + b_\lambda s^\lambda + b_{\lambda+1} s^{\lambda+1} + \cdots + b_\mu s^\mu$$

$$\therefore (b_0 - a_0) + (b_1 - a_1)s + (b_2 - a_2)s^2 + \cdots + (b_\lambda - a_\lambda)s^\lambda + b_{\lambda+1}s^{\lambda+1} + \cdots + b_\mu s^\mu = 0$$

$$\therefore b_0 = a_0, b_1 = a_1, b_2 = a_2, \dots, b_\lambda = a_\lambda, b_{\lambda+1} = 0, \dots, b_\mu = 0$$

綜上，唯一性得证。

